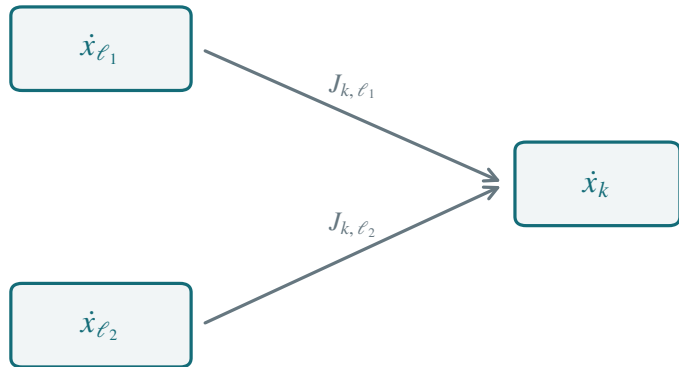
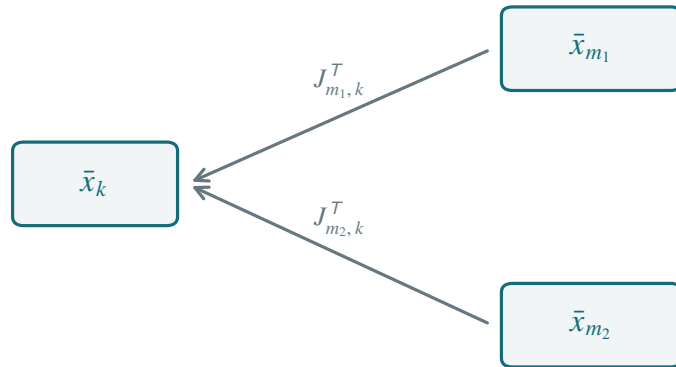


Forward: sum contributions of
parents



$$\dot{x}_k = \sum_{\ell} J_{k\ell} \dot{x}_{\ell}$$

Reverse: sum contributions of
children



$$\bar{x}_k = \sum_m J_{mk}^T \bar{x}_m$$